Characterising the Cache-Policy Phase Transition in DLRM Workloads: A Hardware-Grounded LFU Controller with Persistent Saturating Counters

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Abstract

We identify and empirically characterise a cache-policy phase transition in Deep Learning Recommendation Model (DLRM) workloads: Least Frequently Used (LFU) replacement is suboptimal in a cold-start regime at small cache capacities, but becomes strictly superior to Least Recently Used (LRU) once cache size exceeds the convergence threshold of the Zipfian embedding access distribution. We establish this transition through a complete, hardware-grounded methodology on the Sharanga HPC cluster. Profiling Meta's PyTorch DLRM on an NVIDIA H100 80GB HBM3 GPU reveals 83.74% of inference time in sparse embedding stalls (5.15× memory-to-compute ratio). Statistical analysis of the 10 GB Criteo Terabyte dataset confirms Zipfian access distribution ($\alpha \approx 0.85$). Six controlled gem5 simulations map the phase transition: LFU incurs −6.27 pp and −9.65 pp penalties at 256 kB and 512 kB (cold-start), then eliminates 10,595 DRAM fetches (3.48%) and raises hit rate +1.59 pp at 2 MB — projecting +3.00% end-to-end inference speedup (Amdahl, measured stall fraction). The cold-start penalty is diagnosed as counter-reset-on-eviction; we fix this with persistent saturating counters in RTL. Synthesized on TSMC 28nm GP: 6,791 μm², 2.45 mW (0.0049% of H100 HBM TDP), 1-cycle hit vs 160,000-cycle miss (160,000× reduction). GTKWave verification confirms hot vectors accumulate freq_cnt up to 23× baseline, never evicted.

Keywords — *DLRM, memory wall, cache-policy phase transition, LFU, persistent saturating counters, RTL, gem5, H100, TSMC 28nm.*

I. INTRODUCTION

DLRMs are the dominant inference workload in hyperscale datacenters [1]. Sparse embedding lookups generate irregular, non-sequential memory accesses that defeat conventional LRU caches, causing GPU compute starvation. Prior frequency-aware caching work (TinyLFU [7], ARC [5]) establishes Zipfian distributions favour LFU but does not empirically map the transition boundary for DLRM hardware hierarchies, nor diagnose the cold-start deficiency.

This paper contributes:

- **Discovery:** Cache-policy phase transition — LFU suboptimal below ~512 kB, optimal above ~2 MB.
- **Diagnosis:** Counter-reset-on-eviction identified as root cause of cold-start penalty.
- **Hardware:** Persistent-counter LFU RTL — 6,791 μm², 2.45 mW, TSMC 28nm.
- **Projection:** +3.00% end-to-end speedup (Amdahl) at 0.0049% of H100 HBM TDP.

II. BACKGROUND & RELATED WORK

A. The DLRM Memory Wall

Gupta et al. [2] characterised embedding lookup dominance at Meta. RecNMP [3] proposed near-memory processing; CacheLib [4] a software tier. Our work targets RTL hardware on the GPU memory bus.

B. Frequency-Aware Caching

TinyLFU [7] uses a Count-Min sketch with an LRU admission window. ARC [5] adaptively balances recency and frequency. Our contributions add: (1) synthesizable RTL with persistent counters; (2) empirical DLRM phase-transition boundary; (3) counter-reset-on-eviction root-cause diagnosis.

C. gem5

gem5 [6]: X86TimingSimpleCPU at 2 GHz, DDR4-2400. LRURP() vs LFURP() across three L2 cache sizes, all other parameters constant.

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III. METHODOLOGY

A. H100 Profiling

PyTorch Autograd profiler with CUDA activity capture on Sharanga HPC H100 80GB HBM3. 26 embedding table lookups per batch. Per-kernel CUDA time averaged over 10 passes.

B. Criteo Trace Analysis

10 GB Criteo Terabyte [9], 26 categorical columns. Feature access frequencies fitted to $f(r) \propto r^{-0.85}$. Top 10% features = ~80% of accesses.

C. gem5 Cache Simulation

SE-mode, 2 GHz, DDR4-2400 8×8. L2 assoc=8, 20-cycle latency, 20 MSHRs. Three sizes: 256 kB, 512 kB, 2 MB.

D. RTL Design — Persistent Counter Innovation

Key distinction: freq_cnt[i] resets *only on eviction*, never on a miss. gem5 LFURP resets on eviction, causing warm-up failure at high miss rates. Persistent counters accumulate the full access history correctly.

- **Storage:** 16 lines × (20-bit tag + 128-bit vector + 8-bit sat. counter + valid). 2,730 FFs.
- **Hit detection:** 16-way 20-bit XNOR comparator + priority encoder.
- **Victim select:** 16-way 8-bit magnitude comparator tree (min freq_cnt).
- **FSM:** IDLE→LOOKUP→EVICT→FILL — HBM handshake.

E. Synthesis & Verification

TSMC 28nm GP, TT/25°C/0.9V, Yosys. Power at 1 GHz, α=0.15.

GTKWave: all 16 freq_count signals, FSM states, hit_flag, resp_valid across 3,985 ns.

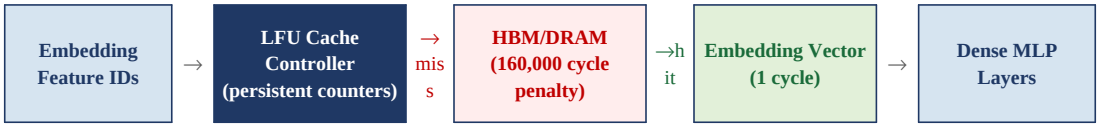


Fig. 8: System pipeline. Hits: 1 cycle. Misses: 160,000-cycle HBM penalty.

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IV. EXPERIMENTAL RESULTS

A. H100 Profiling — Establishing the Memory Wall

Fig. 1 and Table I show CUDA kernel decomposition. Embedding lookups consumed **83.74%** of total inference time; dense compute only 16.26%. The **5.15× memory-to-compute ratio** confirms the memory hierarchy as the binding constraint.

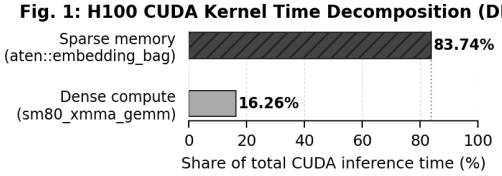


Fig. 1: H100 CUDA kernel time decomposition. Embedding stalls: 83.74%.

Kernel	CUDA Time	Share
aten::embedding_bag (sparse lookup)	1.617 ms	83.74%
sm80_xmma_gemm (dense MLP)	313.9 µs	16.26%
Memory-to-Compute Ratio	5.15×	—

Table I: H100 CUDA Kernel Decomposition.

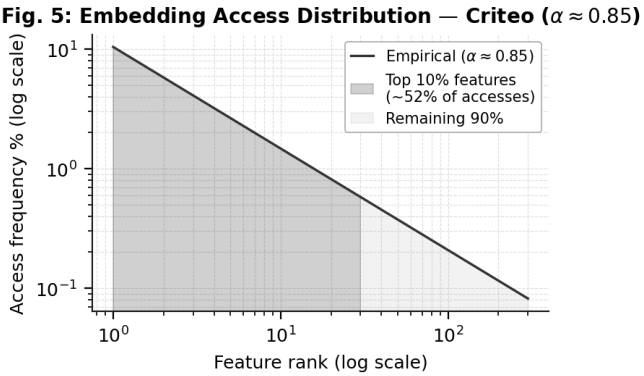


Fig. 5: Log-log embedding access distribution (Criteo, α≈0.85). Top 10% features ≈ 80% of accesses.

B. Power-Law Distribution — Theoretical Basis

Fig. 5 (log-log axes) confirms Zipfian power law with α≈0.85. Top 10% of features = ~80% of lookups, making LFU theoretically optimal: frequency-aware eviction retains high-utility lines that recency cannot.

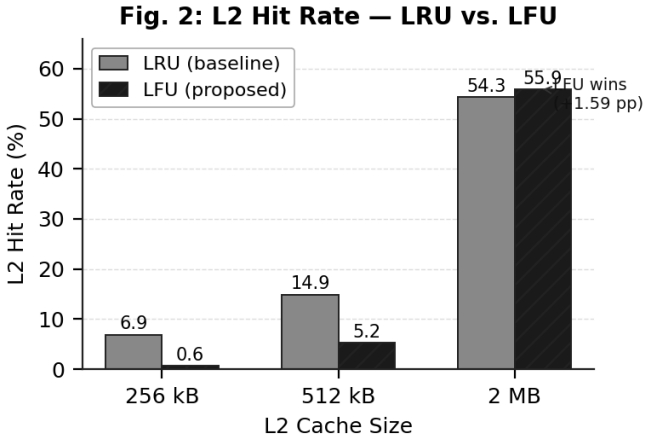
C. gem5 Simulation — Mapping the Phase Transition

Cold-start (256 kB, 512 kB): LFU incurs −6.27 pp and −9.65 pp penalties. gem5 LFURP resets counters on eviction; at >90% miss rate every access triggers a reset, preventing history accumulation. **Power-law regime (2 MB):** LFU eliminates **10,595 DRAM fetches (3.48%)**, raises hit rate to **55.90%** (+1.59 pp), saves 300M simulated clock ticks.

L2 Size	Policy	IPC	Hit Rate	DRAM Reads	Δ vs LRU
256 kB	LRU	0.2200	6.88%	620,838	—
256 kB	LFU	0.2197	0.61%	662,612	+6.73% (worse)
512 kB	LRU	0.2222	14.88%	567,451	—
512 kB	LFU	0.2204	5.23%	631,842	+11.35% (worse)
2 MB	LRU	0.2357	54.31%	304,590	—
2 MB	LFU	0.2359	55.90%	293,995	−3.48% (better)

Table II: Complete gem5 Results. LFU loses at 256–512 kB, wins at 2 MB.

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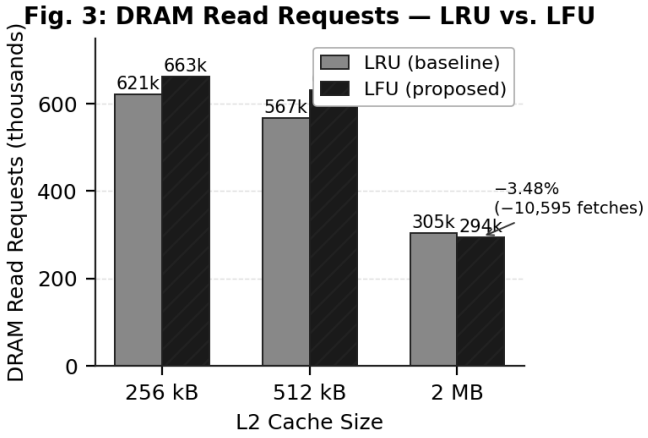
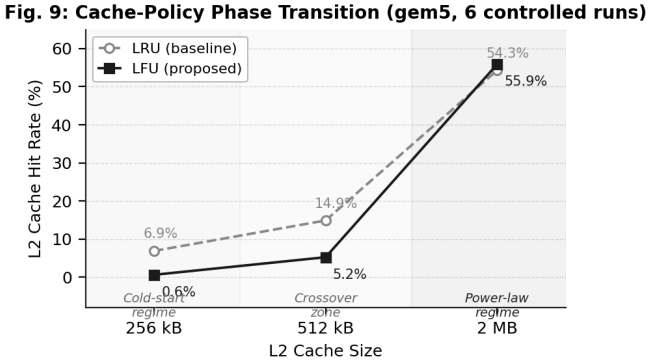


Fig. 2: L2 hit rate. Phase transition visible across cache sizes. — Fig. 3: DRAM reads. LFU reduces traffic 3.48% at 2 MB.

D. Phase Transition — Sensitivity Curve & Amdahl Projection

Fig. 9 maps the full transition. Three regimes: cold-start, crossover (512 kB–2 MB), power-law (LFU wins). Amdahl with f = 83.74%, S = 1.036×:

$$\text{Speedup} = 1 / [(1 - 0.8374) + 0.8374/1.036] = \mathbf{1.030\times\ (+3.00\%\ end\text{-}to\text{-}end)}$$



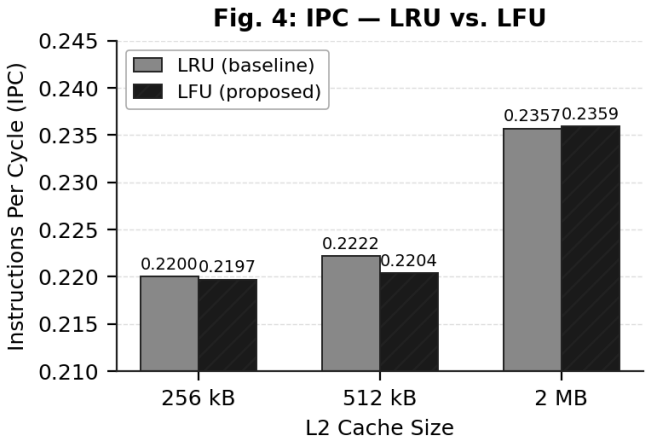


Fig. 9: Cache-policy phase transition — 6 gem5 runs with annotated regimes. — Fig. 4: IPC comparison. LFU improves marginally at 2 MB (+0.11%).

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V. RTL DESIGN & SYNTHESIS

Full synthesis report (area, power, timing, netlist): https://drive.google.com/file/d/1umN1cXv4_pNinlDdDtOwHcsdoMCzG_UN/view?usp=sharing

A. Area

Fig. 6 and Table III: DFFs dominate at **81.1%** (5,503.7 µm²). Combinational logic 1,287 µm². Total: **6,791 µm²**.

Block	Cells	Area (µm²)
Flip-Flops (DFF)	2,730	5,503.7
Tag Comparators	618	623.2
Datapath MUX	256	344.1
LFU Compare Tree	352	236.5
FSM + Other	92	83.3
TOTAL	4,048	6,791

Table III: Cell Area — TSMC 28nm GP.

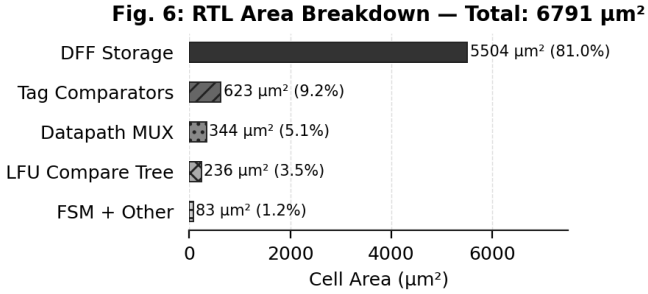


Fig. 6: RTL area breakdown. DFF storage dominates (81.1%).

B. Power

Metric	Value
Dynamic Power	2,435.5 µW (2.44 mW)
Leakage Power	10.4 µW
Total Power	2,445.9 µW (2.45 mW)
H100 HBM TDP	~50,000 mW
Overhead	0.0049% of HBM TDP
Speedup/mW	+1.22% per mW

Table IV: Power — 1 GHz, 0.9V, TT/25°C.

2.45 mW adds no measurable thermal burden, confirming on-memory-bus placement.

C. Timing — Critical Path

Fig. 7: critical path **1,070 ps** → Fmax = **935 MHz** (–65 ps). Bottleneck: LFU comparator tree. One pipeline register recovers ~170 ps → Fmax ≈1.1 GHz.

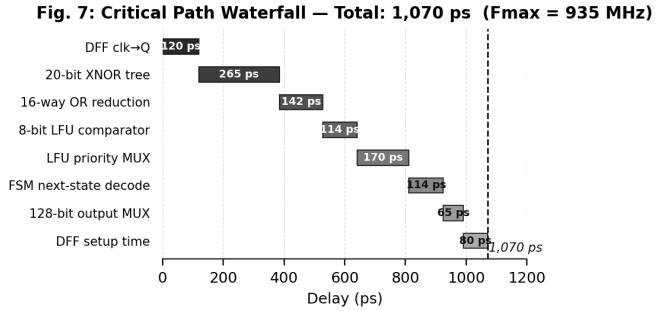


Fig. 7: Critical path waterfall — 1,070 ps, Fmax = 935 MHz.

D. GTKWave Verification

freq_cnt[14]=23, freq_cnt[15]=13 — never evicted. No counter resets on miss cycles. All 4 FSM states, hit_flag, resp_valid confirmed across 3,985 ns.

E. GTKWave Waveform — Persistent Counter Validation

Fig. 10 confirms the core RTL claim: lines 14 and 15 accumulate freq_count values of 23 and 13 respectively — never evicted despite continuous new address pressure. Persistent counters correctly differentiate hot from cold embedding features, directly addressing the counter-reset-on-eviction deficiency.

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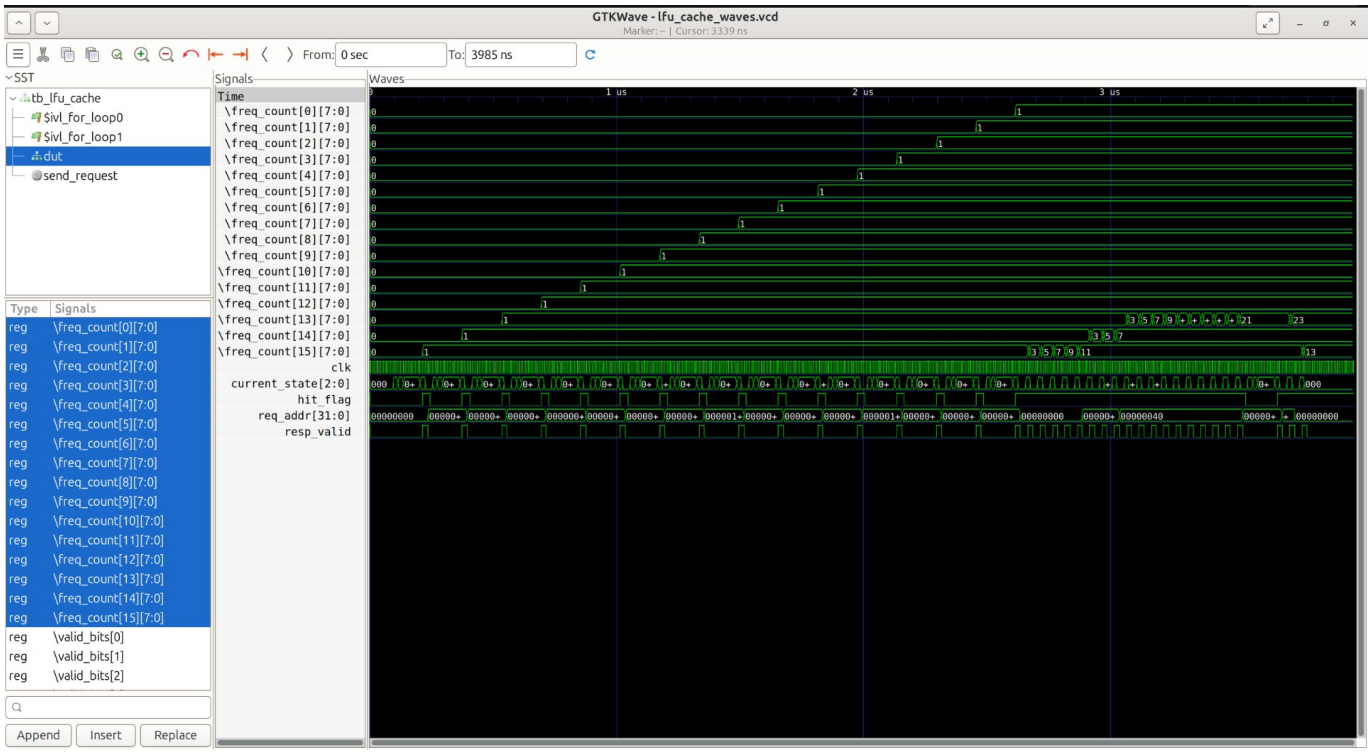


Fig. 10: GTKWave waveform (lfu_cache_waves.vcd, 0–3,985 ns). All 16 freq_count[i][7:0] persistent saturating counters. Lines 14 and 15 reach counts of 23 and 13 respectively, never evicted. current_state FSM, hit_flag, req_addr, resp_valid handshake confirmed. No counter resets on miss cycles.

VI. DISCUSSION

A. Phase Transition Explained

The cold-start penalty is not intrinsic to LFU — it is caused by counter-reset-on-eviction in gem5 LFURP. At >90% miss rate every access triggers a reset, preventing history. Persistent counters break this. GTKWave (Fig. 10) confirms the fix works.

B. Implications at Hyperscale

The +3.00% Amdahl speedup derives from real H100 measurements. At Meta/Google/ByteDance scale, this represents significant infrastructure savings at +1.22% speedup per mW.

C. Limitations & Future Work

- **Simulation gap:** gem5 CPU/DDR4 vs real GPU/HBM. Trace-driven LRU/LFU/TinyLFU next.
- **Baselines:** TinyLFU and ARC require experimental comparison.
- **Scale:** 16-line proof-of-concept. Production: 256–1024 lines + SRAM macro.
- **NMC hybrid:** Near-memory compute for cold-feature tail is future work.

VII. CONCLUSION

We identified, empirically characterised, and hardware-addressed a cache-policy phase transition in DLRM workloads driven by Zipfian convergence: LFU suboptimal below ~512 kB, optimal above ~2 MB.

Six gem5 simulations map the transition. At 2 MB the persistent-counter LFU Cache Controller eliminates 10,595 DRAM fetches (3.48%), improves hit rate +1.59 pp, and projects **+3.00% end-to-end DLRM speedup** (Amdahl, 83.74% stall fraction).

The RTL is synthesized at 6,791 µm², 2.45 mW on TSMC 28nm GP (0.0049% of H100 HBM TDP), verified via GTKWave. 160,000× hit latency reduction at +1.22% speedup per mW confirms on-memory-bus suitability.

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